

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Additional guidance                                                                                                                                                                                                  | Mark                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 6(a)            | <p>An explanation that includes three of the following points:</p> <ul style="list-style-type: none"> <li>ions are {charged / polar / hydrophilic} and cannot (diffuse / pass / travel) through {phospholipid bilayer / hydrophobic membranes / non-polar membranes} (1)</li> <li>(therefore cross the cell membrane) using {channel / carrier} proteins (1)</li> <li>by {facilitated diffusion / active transport} (1)</li> <li>correct reference to concentration gradient (1)</li> </ul> | <p>allow voltage-gated channels / sodium potassium pump</p> <p>ignore simple diffusion<br/>accept actively {pumped / moved}</p> <p>e.g. ions are pumped against the concentration gradient (in active transport)</p> | <p>Expert</p> <p>(3)</p> |

| Number  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| * 6 (b) | <p>Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme.</p> <p>The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. (If in doubt about role PLEASE CHECK)</p>                               |
| 1       | <ul style="list-style-type: none"> <li>consideration of concentration or distribution of one ion using the graph /data</li> </ul>                                                                                                                                                                                                                                                                                                                                                           |
| 2       | <ul style="list-style-type: none"> <li>consideration of concentration or distribution of two or more ions using the graph / data</li> </ul>                                                                                                                                                                                                                                                                                                                                                 |
| 3,      | <ul style="list-style-type: none"> <li>consideration of role of <math>\text{Na}^+</math> in the human body ( linked to comment about relative concentration)</li> <li>role in muscle contraction /role in nerve impulse /role in water balance as an electrolyte</li> </ul> <p>accept any correct stated role for <math>\text{Na}^+</math></p>                                                                                                                                              |
| 4       | <ul style="list-style-type: none"> <li>consideration of role of <math>\text{K}^+</math> in the human body ( linked to comment about relative concentration)</li> <li>role in nerve impulse / role in muscle contraction / role in active transport / role in protein synthesis / role in activation of some enzymes</li> </ul> <p>accept any correct stated role for <math>\text{K}^+</math></p>                                                                                            |
| 5       | <ul style="list-style-type: none"> <li>consideration of role of <math>\text{Cl}^-</math> in the human body ( linked to comment about relative concentration)</li> <li>role in body fluid balance / osmotic regulation / role in chloride shift / role in production of HCl in stomach / role in neuronal hyperpolarisation /role in activation of enzymes / role in resting potential maintenance in neurones</li> </ul> <p>accept any correct stated role for <math>\text{Cl}^-</math></p> |
| 6       | <ul style="list-style-type: none"> <li>consideration of role of <math>\text{HCO}_3^-</math> in the human body ( linked to comment about relative concentration)</li> <li>role in pH buffering / role in carbon dioxide transport in blood</li> </ul> <p>accept any correct stated role for <math>\text{HCO}_3^-</math></p>                                                                                                                                                                  |

|         |     |                                                                                                                                                                                                                                                                                                                                     | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Level 0 | 0   | No awardable content                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Level 1 | 1-2 | <p>An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information.</p> <p>The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.</p>       | <p>limited consideration of two or of the following using data from the graph.</p> <p>eg.<br/> <math>\text{Na}^+</math> higher in tissue fluid and blood plasma than cytoplasm (with comparative figures)</p> <p><math>\text{Na}^+</math> lower in cytoplasm than <math>\text{K}^+</math><br/> <math>\text{Na}^+</math> _ higher in tissue fluid and blood plasma than <math>\text{K}^+</math><br/>           -</p> |
| Level 2 | 3-4 | <p>An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information.</p> <p>The explanation shows some linkages and lines of scientific reasoning, with some structure.</p>                                                                            | <p>limited consideration of roles of two ions in the human body.</p> <p>If only one ion max 3</p>                                                                                                                                                                                                                                                                                                                   |
| Level 3 | 5-6 | <p>An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information.</p> <p>The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.</p> | <p>detailed science behind role of two ions in the human body</p> <p>If only one ion max 5</p>                                                                                                                                                                                                                                                                                                                      |

